

Quiz 1 (Version 1)

CAS CS 132: *Geometric Algorithms*

September 21, 2026

Name:

BUID:

- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Row Operations

Apply the row operations:

$$R_3 \leftarrow R_3 + 2R_2$$

$$R_1 \leftrightarrow R_2$$

$$R_3 \leftarrow -2R_3$$

from top to bottom to the following matrix. You must write down the intermediate matrices and row operations you used in your calculation.

$$\begin{bmatrix} 4 & 5 & -1 \\ 2 & -9 & -3 \\ -3 & 5 & -2 \end{bmatrix}$$

Solution.

2 Reduced Echelon Froms

Determine the reduced echelon form of the following matrix. You must write down the intermediate matrices and row operations you used in your calculation.

$$\begin{bmatrix} 2 & 4 & -6 & 0 \\ -3 & -6 & 9 & 3 \\ -1 & -2 & 3 & -6 \end{bmatrix}$$

Solution.

3 General Form Solutions

Determine a general form solution for a linear system whose augmented matrix is row equivalent to the following matrix (*Hint*. First determine if the following matrix is in reduced echelon form).

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 5 & 3 \\ 0 & 1 & 1 & 1 & -1 & 4 \\ 0 & 0 & 1 & 3 & -4 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.